

Linear Time-Delay Control by Shaping Delay Distribution

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I. INTRODUCTION

Time delays play a critical role in numerous systems across engineering, economics, and life sciences, as the transfer of material, energy, and information is rarely instantaneous [1]. Delays arise in many settings, including computational and communication latencies, transport and propagation phenomena, hereditary effects in population dynamics, and delayed feedback in control systems. They can be intrinsic to the underlying dynamics or introduced through modeling approximations of more complex distributed parameter systems [2]. Time delays make control more challenging because delayed feedback prevents the controller from reacting to the current system state, often leading to performance degradation or instability. In contrast, numerous examples are also known in which delays improve stability [3].

We consider Time-Delay Systems modeled by functional differential equations. The simplest representative is a delay-differential equation with a single discrete delay. For this instance, the state trajectory $t \rightarrow x(t) \in \mathbb{R}^n$ will be determined by

$$\dot{x}(t) = F(x(t), x(t - \tau)), \quad F : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n. \quad (1)$$

In this instance, the trajectory is determined by the current and the past state at one discrete point in time. Delay-differential equations can also exhibit distributed delay (2), in which the system's evolution depends on a weighted average of its past states. These kinds of delays have been shown to model biological phenomena better than discrete delays [4].

$$\dot{x}(t) = \int_{-\tau}^0 g(\theta) x(t + \theta) d\theta, \quad g(\cdot) : [-\tau, 0] \rightarrow \mathbb{R}^n \times \mathbb{R}^n. \quad (2)$$

II. CONTROLLER DESIGN

The stability analysis framework in [3] has been extended for systems of the form

$$\begin{aligned} \dot{x}(t) &= \sum_{k=0}^m A_k x(t - \tau_k) + \sum_{k=0}^m B_k u(t - \tau_k) + \int_{-\tau_I}^0 G(\theta) x(t + \theta) d\theta \\ y(t) &= \sum_{k=0}^m C_k x(t - \tau_k). \end{aligned} \quad (3)$$

for which $A_k \in \mathbb{R}^{n \times n}$, $B_k \in \mathbb{R}^{n \times n_u}$, $C_k \in \mathbb{R}^{n_y \times n}$, $G(\cdot) : [-\tau, 0] \rightarrow \mathbb{R}^{n \times n}$ and $\tau_0 = 0 \leq \tau_1 \leq \dots \leq \tau_m < \infty$.

The framework makes it possible to synthesize static output feedback controllers $u(t) = K y(t)$ stabilizing systems of the form (3) with a real valued matrix K . Another possibility is to design a dynamic output feedback controllers with a chosen order n_c

$$\begin{cases} \dot{x}_c(t) = A_c x_c(t) + B_c y(t) \\ u(t) = C_c x_c(t) + D_c y(t) \end{cases} \quad (4)$$

for which $x_c(t) \in \mathbb{R}^{n_c}$, and A_c, B_c, C_c , and D_c are real-valued matrices with appropriate dimensions. The controller design algorithm is concerned with minimizing the spectral abscissa, i.e., the real part of the right-most characteristic root, with respect to the gain matrix K or the elements of the matrices in (4). Since a strictly negative spectral abscissa is both a necessary and sufficient condition for stability, the proposed design method is non-conservative. Consequently, a stabilizing controller can be computed whenever one exists. The spectral abscissa is usually non-convex and non-smooth, necessitating dedicated optimization algorithms. To obtain a satisfactory solution, it may be necessary to sample different initial values. The framework can be extended to optimize robustness and performance criteria expressed in terms of the $\mathcal{H}_2$ and $\mathcal{H}_\infty$ criteria.

III. SHAPING THE DELAY DISTRIBUTION

Under earlier assumptions, it was also made possible to construct control laws using distributed delays. This means that the framework is not only able to stabilize systems with distributed delays but also can employ distributed delay output feedback to stabilize general Time-Delay Systems. A strong argument for the inclusion of distributed delays in the control loop is that for Time-Delay Systems the natural state is a function segment. Additionally, distributed delays arise naturally when developing predictor-based control strategies. Hence, a controller incorporating both static feedback and a distributed delay is designed as

$$\begin{aligned} u(t) &= K_1 y(t) + K_2 \int_{-\tau_I^C}^0 g(\theta) y(t + \theta) d\theta \\ g(\theta) &= \sum_{k=0}^K c_k T_k \left(2 \frac{\theta}{\tau_I^C} + 1 \right) \end{aligned} \quad (5)$$

with the optimization variables $K_1, K_2 \in \mathbb{R}^{n_y \times n_u}$ and $c_k \in \mathbb{R}$. T_k denotes the Chebyshev polynomial of the first kind of degree k and $\tau_I^C > 0$ represents the integration horizon of the distributed delay.

REFERENCES

- [1] T. Erneux, *Applied delay differential equations*, 1 2009. [Online]. Available: <https://doi.org/10.1007/978-0-387-74372-1>
- [2] W. Michiels, *Theory of Control of Linear Time-Delay Systems*, 1 2024. [Online]. Available: <https://doi.org/10.1016/b978-0-443-14081-5.00032-5>
- [3] P. Appeltans, H. Silm, and W. Michiels, "TDS-CONTROL: a MATLAB package for the analysis and controller-design of time-delay systems*," *IFAC-PapersOnLine*, vol. 55, no. 16, pp. 272–277, 1 2022. [Online]. Available: <https://doi.org/10.1016/j.ifacol.2022.09.036>
- [4] R. M. May, "Time-Delay Versus Stability in Population Models with Two and Three Trophic Levels," *Ecology*, vol. 54, no. 2, pp. 315–325, 3 1973. [Online]. Available: <https://doi.org/10.2307/1934339>